

Sudarshan Pro

A phone-based remote control for Windows PCs and smart boards — designed for classrooms, lecture halls and institutions.
One small Windows executable. No installation wizard. No cables. No cloud.

Developer: Jai Servana Bhava **Roll No:** 26B25A4614
Distribution: Single Windows executable (`SudarshanPro.exe`)
Audience: Lecturers, institutions, IT administrators

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1. Executive Summary

Sudarshan Pro turns any smartphone into a full remote control for a Windows PC or smart board — over the institution's own Wi-Fi network. A lecturer keeps their phone in their pocket, walks up to the board, and controls *everything* from it: the mouse cursor, keyboard, volume, brightness, monitor power, media playback, files, screenshots, the webcam, and even a live view of the PC screen.

The product is delivered as a **single Windows executable**. Double-click it, a QR code appears on the PC, the lecturer scans it once with their phone, and the connection is ready. There is nothing to install on the phone — the entire remote interface runs in the phone's web browser.

In one line: Sudarshan Pro removes the walk between the podium and the PC, and replaces a chain of five or six awkward steps with a single tap.

2. The Problem — A Day in a Lecturer's Life

"Prof. Rao is explaining a diagram on the smart board when a student asks him to share the notes. The notes are on his phone. He walks back to the PC, wakes it, connects it to the institution Wi-Fi, opens WhatsApp Web, scans the QR code with his phone, sends the file to himself, downloads it on the PC, and walks back to the board. Four minutes of the lecture are gone — and the class has lost the thread of the explanation."

Smart boards have genuinely transformed teaching — lessons are visual, interactive and engaging. But the **PC behind the board** is still controlled the old way: someone must physically stand at it. Lecturers across institutions quietly juggle the same small frictions every single day. None of these are anyone's fault — they are simply what happens when a touch-first teaching style meets a desktop-first operating system:

2.1 Everyday friction points lecturers face

Situation	What the lecturer has to do today
Send a photo / document from phone to the board PC	Walk to the PC → connect PC to Wi-Fi → open WhatsApp Web (or e-mail/drive) → scan QR → send the file to themselves → download on PC → open it. Five to six steps, every time.
Move the cursor while standing at the board	Reach for the wireless mouse/keyboard kept "somewhere near the podium", or walk back to the desk. The cursor also vanishes on touch boards after pen input.
Type a URL or a quick search mid-lecture	Walk back and type on the physical keyboard, or hunt for the on-screen keyboard.
Adjust volume or screen brightness	Locate the physical buttons — which differ on every board model — or navigate Windows settings on the PC itself.
Blank the screen during a discussion	Walk to the PC to lock it, or let the lesson stay projected while talking.
Projector / monitor sleeps mid-explanation	Wiggle the physical mouse or press a keyboard key — the display wake never happens from the front of the hall.
See what's actually on the PC from the back bench	Physically check the screen; there is no way to glance at the PC display on the phone.
A student app or pop-up steals focus	Alt-Tab or Task Manager from the PC — walking back again.
Collect / distribute files quickly	Pass a pen drive around, or ask students to e-mail files that then land "somewhere" in the inbox.
Quickly open an installed application	Minimize everything, find the icon on the desktop or Start menu, from the PC.

2.2 Why existing habits don't fully solve it

- **Messengers on the lecture PC** mean personal notifications can pop up on the projector — something every lecturer wants to avoid in front of a class.
- **USB pen drives** are slow to pass around, easy to lose, and a hygiene/security concern on shared machines.
- **Specialized presentation remotes** cost money, need batteries, pair via USB, and usually offer only "next slide" — not mouse, keyboard, files or the screen itself.
- **Cloud drives** need the PC signed in, sync time, and institutional logins — and still require walking to the PC.

The root cause is simple: the phone — the one device every lecturer already carries — has no safe, quick bridge to the board PC. The problem is not the lecturer's habit; it is the missing bridge. **Sudarshan Pro is that bridge.**

3. The Proposed Solution

Sudarshan Pro is a lightweight Windows application that runs quietly on the board PC and exposes a secure web interface to the lecturer's phone over the local Wi-Fi network. The phone becomes a **touchpad, keyboard, media remote, file courier, screen viewer and system panel** — all in one, with no app to install on the phone.

3.1 The same moment, re-imagined

"Prof. Rao taps  Upload to PC on his phone, picks the notes PDF, and they're already in the DropZone folder on the board PC — he opens it without leaving the board. Total time: about ten seconds. The class never loses momentum."

3.2 Solution at a glance

Need	Before (typical today)	With Sudarshan Pro
Phone → PC file transfer	WhatsApp Web / e-mail / pen drive (5–6 steps, walk to PC)	One tap → lands in the DropZone folder on the Desktop
Control the cursor	Find the mouse, or walk back	Phone touchpad with click / right-click / scroll
Type text	Physical keyboard at the PC	Type on the phone, it appears on the PC
Volume / brightness	Hunt for hardware buttons	Dedicated buttons on the phone UI
Blank / wake the display	Walk to PC, press a key	Monitor Off / On buttons; wake is held awake so it won't instantly sleep again
See the PC screen	Physically look at it	Live screen stream + screenshots saved to DropZone
Run a command / open an app	Walk to PC, open Start menu	Built-in app launcher + quick command box
Stop using it	—	Change the PIN, or tap Close on PC from the phone

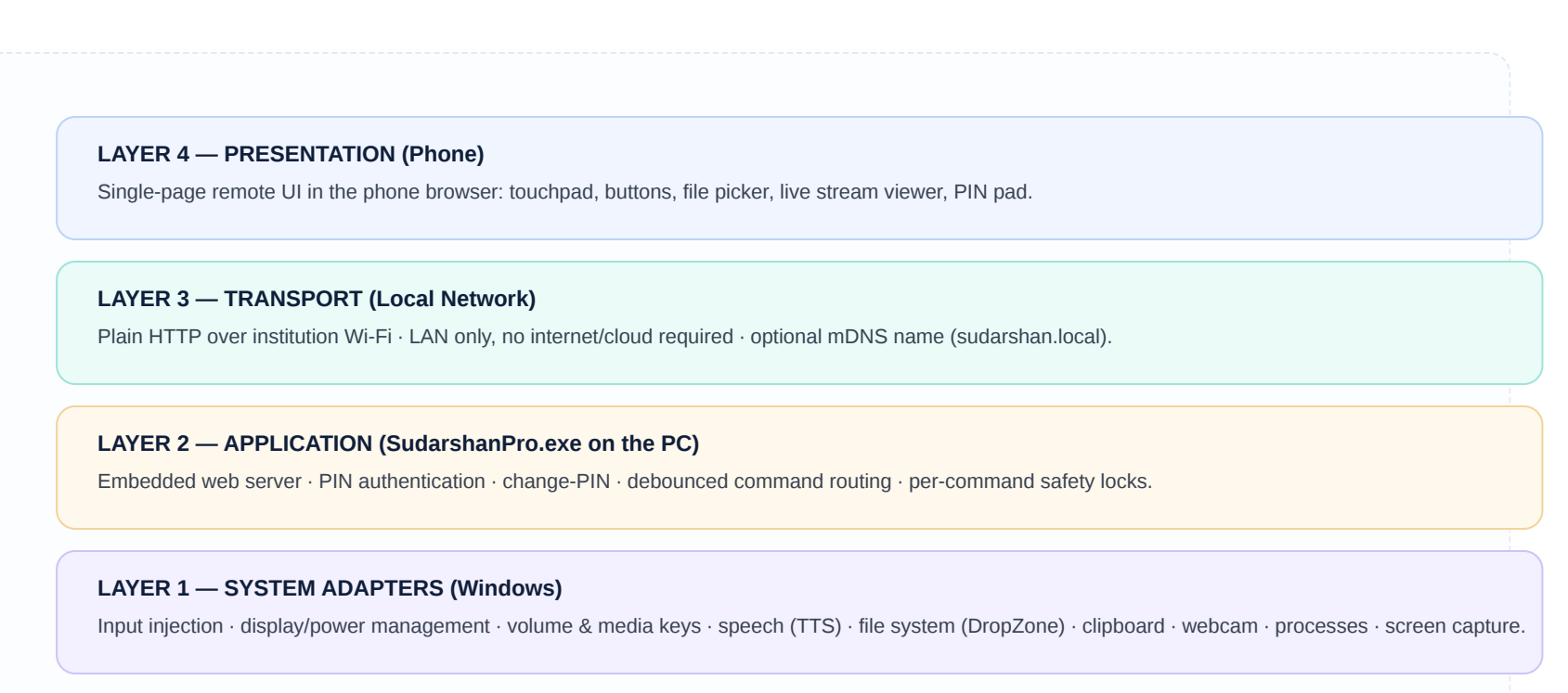
4. How Sudarshan Pro Works

4.1 Connection (one time, ~20 seconds)

1. Run `SudarshanPro.exe` on the board PC. A window with a QR code appears.
2. Connect the **phone to the same Wi-Fi** as the PC (institution network).
3. Scan the QR code — the remote UI opens in the phone browser. (A plain IP address works too.)
4. Enter the PIN. Done — the phone is now the PC's remote.

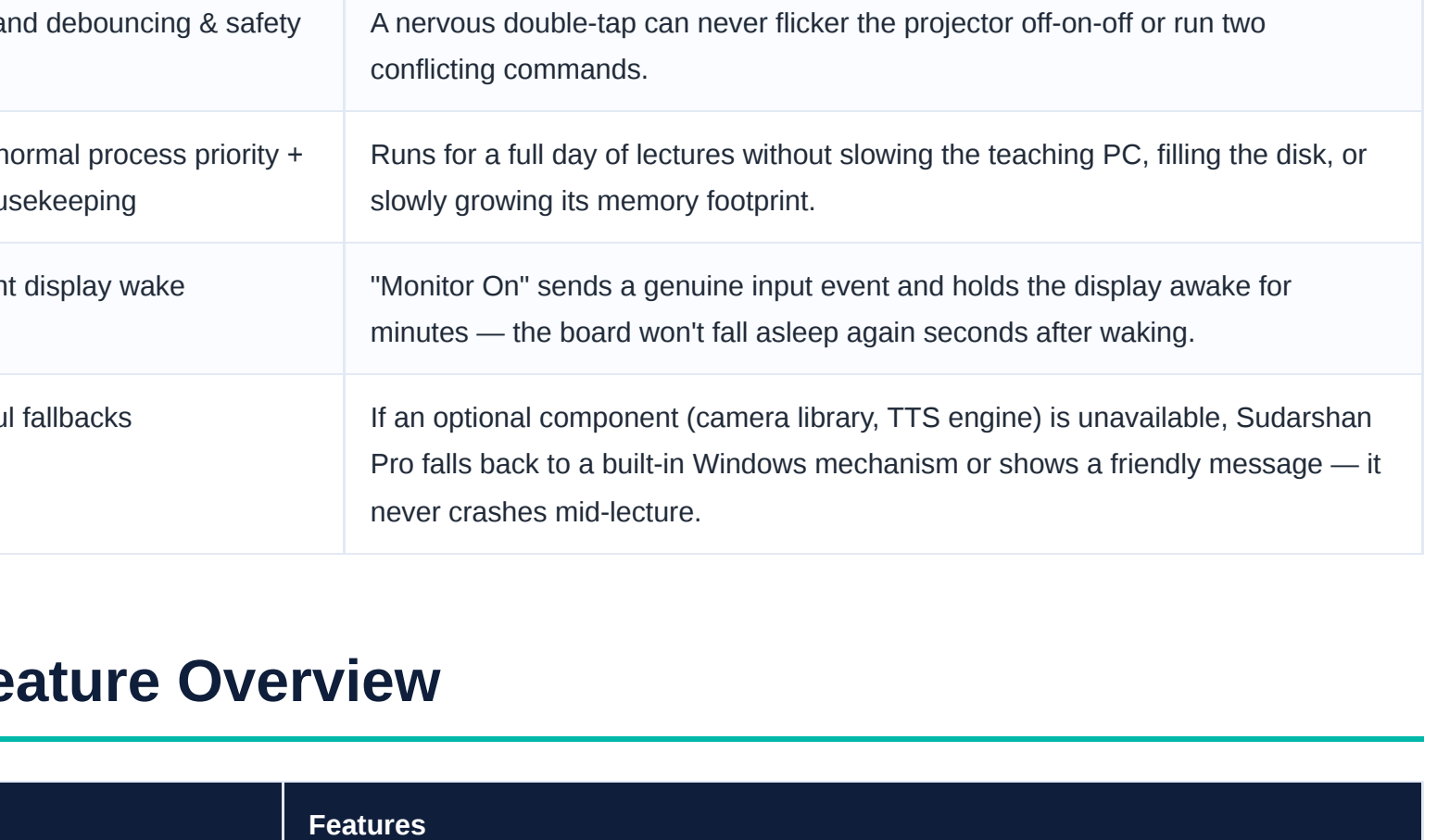
4.2 Everyday use

1. Keep the phone in hand or pocket while teaching at the board.
2. Use the touchpad to point, click and scroll — the cursor is **kept visible** on touch boards automatically.
3. Send files, type text, adjust volume/brightness, control media, stream the screen — all without leaving the board.
4. When class ends: change the PIN or press **Close Sudarshan Pro** from the phone. Nothing keeps running that shouldn't.



5. Architecture

Sudarshan Pro follows a simple, robust **client-server architecture** with four logical layers. Because everything runs on the local network, it keeps working even if the institution's internet connection drops mid-lecture.



5.1 Key architectural decisions

Decision	Why it matters in a classroom
Browser-based phone UI (no app install)	Works on any Android or iPhone the lecturer already owns; nothing for the institution to deploy or update on phones.
Single Windows executable	No installer, no admin rights, no dependencies — IT can place it on the board PC and pin it to startup.
Command debouncing & safety locks	A nervous double-tap can never flicker the projector off-on-off or run two conflicting commands.
Below-normal process priority + self-housekeeping	Runs for a full day of lectures without slowing the teaching PC, filling the disk, or slowly growing its memory footprint.
Resilient display wake	"Monitor On" sends a genuine input event and holds the display awake for minutes — the board won't fall asleep again seconds after waking.
Graceful fallbacks	If an optional component (camera library, TTS engine) is unavailable, Sudarshan Pro falls back to a built-in Windows mechanism or shows a friendly message — it never crashes mid-lecture.

6. Feature Overview

Group	Features
	Multi-touch touchpad (with velocity acceleration), left / right / middle click, dedicated scroll strip, cursor kept visible on touch boards.
	Type text to the PC, pop-up messages, PC text-to-speech announcements, clipboard push (phone → PC) and pull (PC → phone).
	Volume up/down/mute, brightness, play/pause/next/previous media keys, "Now Playing" detection, random wallpaper, live FPS screen stream.
	Upload phone → PC (DropZone), download PC → phone, full file browser with icons & sizes, DropZone auto-listing with one-tap downloads.
	Screenshot to DropZone, webcam snapshot, live screen stream, network speed readout, system status (battery, CPU, RAM) polled live.
	Show desktop, Task Manager, Alt-Tab, lock workstation, sleep, shutdown (with confirmation), monitor off/on, close app, run quick commands, process list with kill, installed-app launcher with search.
	PIN lock on the phone UI, change-PIN (persisted on the PC), one-tap "Close Sudarshan Pro on PC" kill switch.

7. Advantages

For the lecturer

- Never leaves the board — every PC action is one tap away.
- Phone-to-PC sharing without messengers, QR scans or e-mail.
- No personal notifications ever appear on the projector.
- Works with the phone they already carry — zero new hardware.
- Confidence features: cursor always visible, monitor stays awake, confirmations on destructive actions.

For the institution

- Zero cost per classroom beyond the existing Wi-Fi.
- No per-phone software to license, install or update.
- Reduces pen-drive sharing — a common malware vector on shared PCs.
- Standardizes one remote workflow across every board brand.
- A single portable exe — easy for IT to audit, deploy and remove.

For IT administrators

- No server, no cloud account, no internet dependency.
- PIN-gated; lecturer can change it anytime; full close-switch from the phone.
- Self-maintaining: cleans its DropZone, stays low-priority, survives all-day sessions.

Compared to alternatives

- **vs. presentation clickers:** full mouse/keyboard/files/screen — not just slides.
- **vs. WhatsApp Web / cloud drives:** no personal accounts on the lecture PC, no internet needed.
- **vs. USB drives:** instant, hygienic, bidirectional.

8. Security & Privacy

- **Local-only communication:** the phone and PC talk over the institution's LAN. Commands never traverse the internet, and no third-party service is involved.
- **PIN gate:** the phone UI is locked until the correct PIN is entered; the PIN is stored on the PC and can be changed from the phone after re-entering the current one.
- **Debounce, locked commands:** destructive or disruptive actions (lock, monitor, power) ignore accidental double-taps and serialize through a single control channel.
- **Visible & controllable:** the PC shows its own QR/window while serving; the lecturer can terminate the whole application from the phone at any moment.
- **No data retention beyond the room:** transferred files live in a visible **DropZone** folder on the Desktop, are auto-pruned, and are fully under the institution's control.

Deployment note: as with any remote-control tool, Sudarshan Pro should be used on a trusted institutional network. The PIN should be changed from the default before first classroom use, and the app should be closed when the board PC is unattended.

9. Deployment Guide

9.1 What you receive

A single portable Windows executable: `SudarshanPro.exe`. No installer, no external files required.

9.2 First run (one time per PC)

1. Copy `SudarshanPro.exe` onto the board PC (e.g., Desktop or a tools folder) and double-click it.
2. A window appears showing a QR code and the PC's address (`http://<PC-IP>:5000`).
3. On the phone, join the **same Wi-Fi** as the PC, scan the QR code (or type the address in the browser).
4. Enter the PIN, then open  **Change PIN** and set your own.
5. (Optional) Ask IT to add the exe to Startup so it's always ready when the board boots.

9.3 Every lecture

Just launch the exe and scan — or keep the page bookmarked and refresh it. Everything else is taps.

9.4 Requirements

Item	Requirement
Board PC	Windows 10/11, connected to the institution Wi-Fi (Ethernet + shared network also works)
Phone	Any modern smartphone with a browser (Android / iOS)
Network	Phone and PC on the same local network; AP isolation (client isolation) disabled on the Wi-Fi
Optional	Webcam for snapshots; best camera support if OpenCV is present on the build PC

10. Real-World Use Cases

Scenario	How Sudarshan Pro helps
Sharing last-minute notes	Upload the PDF from the phone; it appears in DropZone; open it without leaving the board.
Student's doubt about slide 12	Scroll back with the touchpad, zoom with the keyboard, all while standing beside the student.
Playing a video clip	Push the URL to Chrome, control play/pause/volume from the phone, blank the monitor during the discussion after.
Collecting assignments	Students drop files into DropZone; the lecturer downloads them to the phone or shows them instantly.
Cursor disappeared after pen input	Any touchpad movement restores the cursor automatically — no hunting for settings.
Board went to sleep mid-explanation	One tap — Monitor On — and the display stays awake for the rest of the segment.
Reading a long passage aloud	PC Speak reads the typed text with the PC's voice — useful for language or accessibility sessions.
Quick attendance/activity snapshot	Screenshot or webcam capture straight to DropZone with a timestamped file name.
Emergency: something inappropriate on screen	One tap on Monitor Off or Lock PC — instant, discreet, from anywhere in the room.

11. Limitations & Future Scope

11.1 Current limitations (stated honestly)

- Phone and PC must share the same local network; guest networks with client isolation will block the connection.
- Camera snapshots depend on the webcam being free and, on some boards, benefit from an extra imaging library.
- The live stream is a lightweight JPEG feed (~a few frames per second) — perfect for glancing, not for QR-grade remote desktop.
- Designed for a single controlling phone at a time.

11.2 Future scope

- Multi-user sessions with per-lecturer PINs and audit logs for IT.
- End-to-end encryption layer and certificate pinning for highly sensitive environments.
- Scheduled actions (auto-blank at break time, auto-wake before class).
- Android/iOS companion app with widgets and offline QR pairing history.
- Institution dashboard: board PC health, uptime, and usage statistics.

12. Troubleshooting / FAQ

Issue	Fix
Phone can't open the page / QR won't load	Confirm both devices are on the same Wi-Fi ; disable AP/client isolation on that network; try the IP address shown under the QR instead of the name.
Cursor doesn't move or clicks don't register	Tap the touchpad firmly once; check that the PC isn't showing a UAC/elevated window (Windows blocks simulated input there by design); relaunch the exe.
Monitor turns on then immediately off	Use the built-in Monitor On button — it holds the display awake; also raise Windows' display sleep timeout on the board PC.
Camera capture fails	Close apps using the webcam (Teams, Zoom, browser tabs); ensure a camera is connected; for best results rebuild/bundle with the imaging library noted in the deployment guide.
Where did my uploaded file go?	Desktop → DropZone folder. Use  File Drop Zone on the phone to list and download everything.
How do I stop it completely?	Tap  Close Sudarshan Pro on PC at the bottom of the phone UI, or close its window on the PC.
Port 5000 already in use	The app frees the port automatically on startup; if another tool holds it, close that tool or restart the PC once.

13. Conclusion

Lecturers didn't adopt smart boards to spend their classes walking back to a keyboard. The friction was never about the person — it was about the missing bridge between the phone in their hand and the Windows PC behind the board.

Sudarshan Pro is that bridge: a single portable executable, an on-time QR scan, and a PIN — after which the phone quietly becomes the lecturer's mouse, keyboard, file courier, media remote and second screen for the entire session.

Bottom line: fewer steps, no walking, no personal accounts on the lecture PC, and no new hardware to buy. Teaching stays at the board, where it belongs.